

Initiating Search

November 10, 2025, 11:18 AM

• Search:

CAS Lexicon Search:

Clemmensen reduction - Preferred Concept

Search Tasks

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Returned Reference Results from CAS Lexicon (204)	 References	View Results
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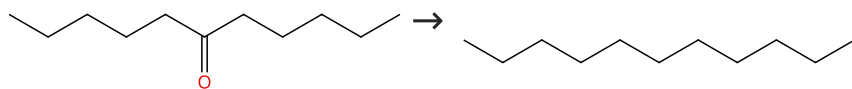
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 98%


 Suppliers (63)

 Suppliers (93)

31-538-CAS-3819059

Steps: 1 Yield: 98%

One-pot reductive cleavage of exo-olefin to methylene with a mild ozonolysis-Clemmensen reduction sequence

- 1.1 **Reagents:** Chlorotrimethylsilane, Zinc
Solvents: Methanol, Dichloromethane; 0.5 h, 0 °C
- 1.2 **Reagents:** Sodium bicarbonate; 5 min, 0 °C

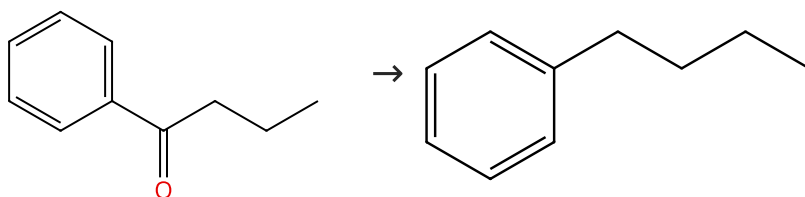
By: Xu, Shu; et al

Tetrahedron Letters (2010), 51(34), 4534-4537.

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 98%


 Suppliers (87)

 Suppliers (68)

31-538-CAS-11614023

Steps: 1 Yield: 98%

Clemmensen reductions using ultrasonic irradiation

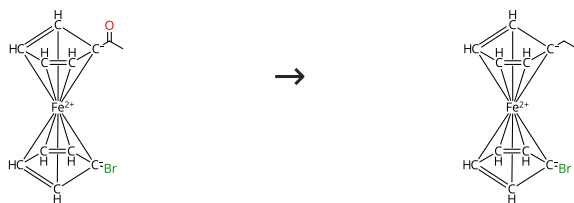
- 1.1 **Reagents:** Hydrogen iodide, Mercury alloy, nonbase, Hg, Zn
Solvents: Methanol, Water

By: Reeves, W. Preston; et al

Synthetic Communications (1988), 18(16-17), 1961-6.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 98%



31-538-CAS-16779135

Steps: 1 Yield: 98%

Synthesis and characterization of 1'-alkylbromoferrocenes

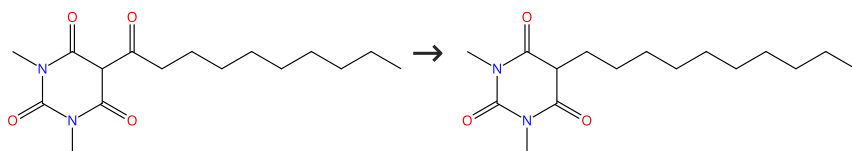
- 1.1 **Reagents:** Zinc, Mercuric chloride, Hydrochloric acid
Solvents: Water
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Methanol, Benzene, Water; reflux; 1.5 h, reflux

By: Bai, Shu-Zhen; et al

Huaxue Shiji (2011), 33(9), 785-787, 790.

Scheme 4 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (2)

Suppliers (2)

31-538-CAS-13526725

Steps: 1 Yield: 97%

Mono C-alkylation and mono C-benzoylation of barbituric acids through zinc/acid reduction of acyl, benzylidene, and alkylidene barbiturate intermediates

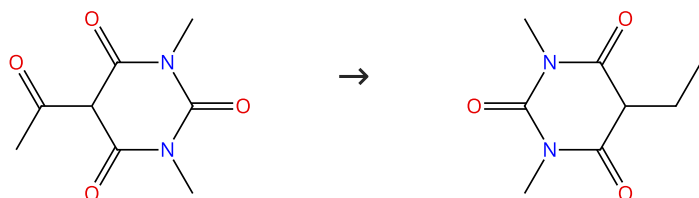
By: Jursic, Branko S.; et al

Tetrahedron Letters (2003), 44(10), 2203-2210.

1.1 **Reagents:** Zinc, Hydrochloric acid
Solvents: Water; 30 min, rt

Scheme 5 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (44)

Suppliers (7)

31-538-CAS-1993934

Steps: 1 Yield: 96%

Mono C-alkylation and mono C-benzoylation of barbituric acids through zinc/acid reduction of acyl, benzylidene, and alkylidene barbiturate intermediates

By: Jursic, Branko S.; et al

Tetrahedron Letters (2003), 44(10), 2203-2210.

1.1 **Reagents:** Zinc, Hydrochloric acid
Solvents: Water; 30 min, rt

Scheme 6 (1 Reaction)

Steps: 1 Yield: 96%



31-538-CAS-16779138

Steps: 1 Yield: 96%

Synthesis and characterization of 1'-alkylbromoferrocenes

By: Bai, Shu-Zhen; et al

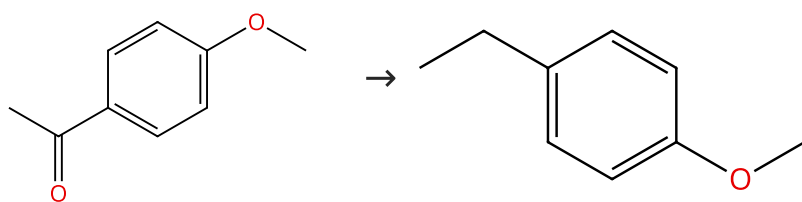
Huaxue Shiji (2011), 33(9), 785-787, 790.

1.1 **Reagents:** Zinc, Mercuric chloride, Hydrochloric acid
Solvents: Water

1.2 **Reagents:** Hydrochloric acid
Solvents: Methanol, Benzene, Water; reflux; 1.5 h, reflux

Scheme 7 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (98)

Suppliers (70)

31-538-CAS-2497544

Steps: 1 Yield: 96%

Clemmensen reductions using ultrasonic irradiation

1.1 Reagents: Hydrogen iodide, Mercury alloy, nonbase, Hg, Zn
Solvents: Methanol, Water

By: Reeves, W. Preston; et al
Synthetic Communications (1988), 18(16-17), 1961-6.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 95%



31-538-CAS-16779139

Steps: 1 Yield: 95%

Synthesis and characterization of 1'-alkylbromoferrocenes

1.1 Reagents: Zinc, Mercuric chloride, Hydrochloric acid
Solvents: Water

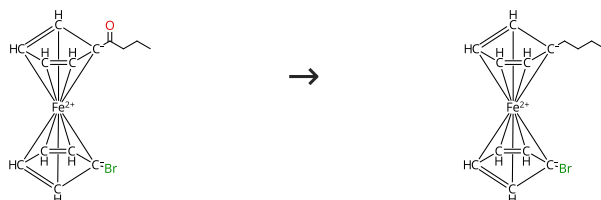
By: Bai, Shu-Zhen; et al

1.2 Reagents: Hydrochloric acid
Solvents: Methanol, Benzene, Water; reflux; 1.5 h, reflux

Huaxue Shiji (2011), 33(9), 785-787, 790.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 95%



31-538-CAS-16779137

Steps: 1 Yield: 95%

Synthesis and characterization of 1'-alkylbromoferrocenes

1.1 Reagents: Zinc, Mercuric chloride, Hydrochloric acid
Solvents: Water

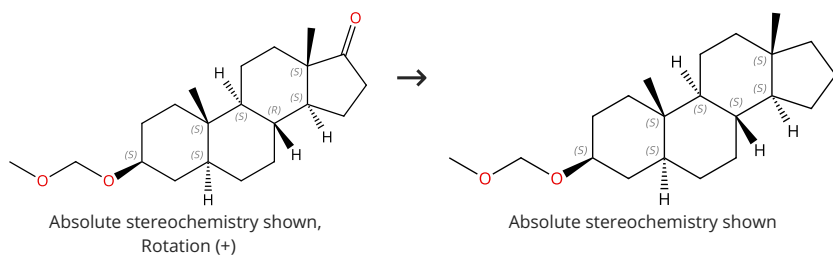
By: Bai, Shu-Zhen; et al

1.2 Reagents: Hydrochloric acid
Solvents: Methanol, Benzene, Water; reflux; 1.5 h, reflux

Huaxue Shiji (2011), 33(9), 785-787, 790.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 94%

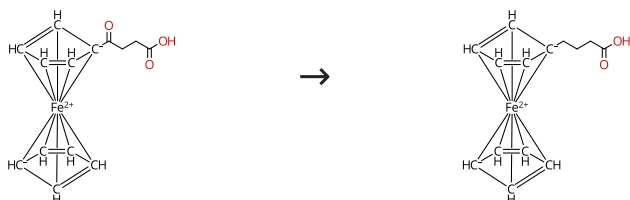


Supplier (1)

31-538-CAS-12963672 Steps: 1 Yield: 94%	One-pot reductive cleavage of exo-olefin to methylene with a mild ozonolysis-Clemmensen reduction sequence
1.1 Reagents: Chlorotrimethylsilane, Zinc Solvents: Methanol, Dichloromethane; 1 h, 0 °C	By: Xu, Shu; et al
1.2 Reagents: Sodium bicarbonate; 5 min, 0 °C Experimental Protocols	Tetrahedron Letters (2010), 51(34), 4534-4537.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (50)

Suppliers (3)

31-538-CAS-11968643 Steps: 1 Yield: 94%	Modified synthesis and electrochemical behavior of two ω-ferrocenyl carboxylic acids
1.1 Reagents: Zinc, Mercuric chloride, Hydrochloric acid Solvents: Water	By: Wang, Peng; et al Huaxue Shiji (2000), 22(4), 196-197.

Scheme 12 (1 Reaction)

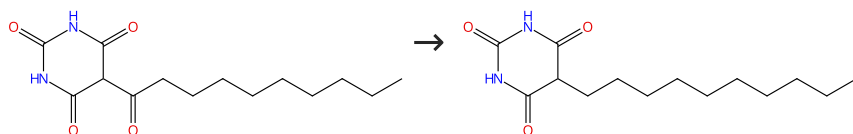
Steps: 1 Yield: 93%



31-538-CAS-16779136 Steps: 1 Yield: 93%	Synthesis and characterization of 1'-alkylbromoferrocenes
1.1 Reagents: Zinc, Mercuric chloride, Hydrochloric acid Solvents: Water 1.2 Reagents: Hydrochloric acid Solvents: Methanol, Benzene, Water; reflux; 1.5 h, reflux	By: Bai, Shu-Zhen; et al Huaxue Shiji (2011), 33(9), 785-787, 790.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 93%



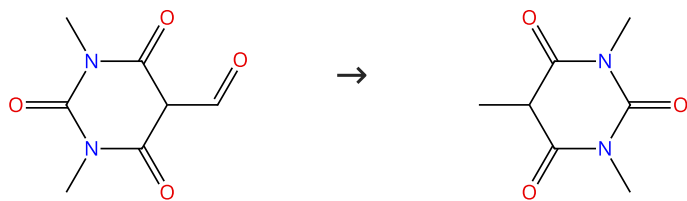
Suppliers (2)

Supplier (1)

31-538-CAS-4135581 Steps: 1 Yield: 93%	Mono C-alkylation and mono C-benylation of barbituric acids through zinc/acid reduction of acyl, benzyldiene, and alkylidene barbiturate intermediates
1.1 Reagents: Zinc, Hydrochloric acid Solvents: Water; 30 min, rt	By: Jursic, Branko S.; et al Tetrahedron Letters (2003), 44(10), 2203-2210.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (24)

Suppliers (11)

31-538-CAS-11121557

Steps: 1 Yield: 91%

Mono C-alkylation and mono C-benylation of barbituric acids through zinc/acid reduction of acyl, benzylidene, and alkylidene barbiturate intermediates

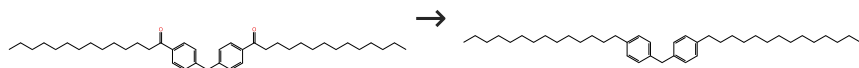
By: Jursic, Branko S.; et al

Tetrahedron Letters (2003), 44(10), 2203-2210.

1.1 Reagents: Zinc, Hydrochloric acid
Solvents: Water; 30 min, rt

Scheme 15 (1 Reaction)

Steps: 1 Yield: 91%



31-538-CAS-4503896

Steps: 1 Yield: 91%

Synthesis and properties of 4,4'-di(n-tetradecyl) diphenyl methane disulfate salt

By: Xiao, Huaying; et al

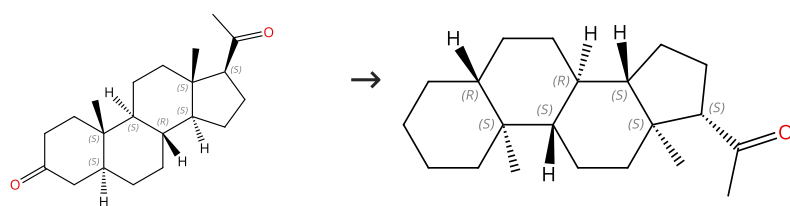
Jingxi Shiyong Huagong (2010), 27(6), 59-62.

1.1 Reagents: Hydrochloric acid
Catalysts: Zinc, Mercuric chloride
Solvents: Water; 5 min, rt

1.2 Reagents: Hydrochloric acid
Solvents: Water; 8 h, reflux; overnight, cooled

Scheme 16 (1 Reaction)

Steps: 1 Yield: 90%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (59)

Suppliers (3)

31-538-CAS-13642688

Steps: 1 Yield: 90%

Ultrasound assisted zinc reactions in synthesis. 2. A new Clemmensen-type reduction

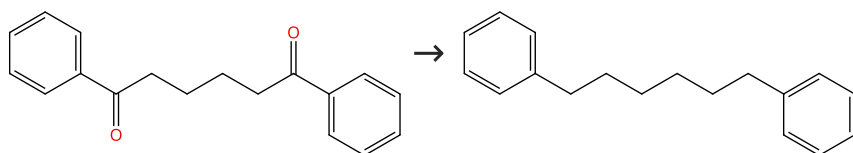
By: Salvador, J. A. R.; et al

Tetrahedron Letters (1993), 34(2), 361-2.

1.1 Reagents: Zinc
Solvents: Acetic acid

Scheme 17 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (59)

Suppliers (22)

31-538-CAS-14048324

Steps: 1 Yield: 90%

Zinc-promoted reactions. 8. The effect of ring strain in the reduction of 1,2-dibenzoylcycloalkanes

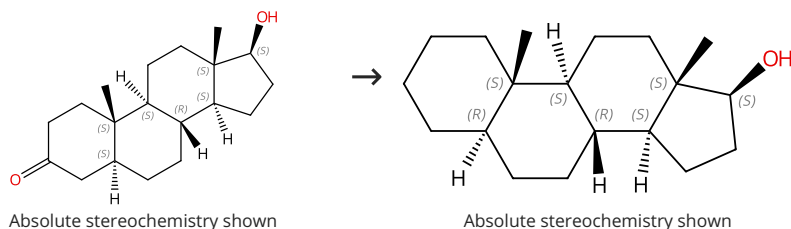
By: Di Vona, Maria Luisa; et al

Tetrahedron (1994), 50(9), 2949-54.

1.1 Reagents: Hydrochloric acid

Scheme 18 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (50)

Suppliers (18)

31-538-CAS-2097372

Steps: 1 Yield: 90%

Ultrasound assisted zinc reactions in synthesis. 2. A new Clemmensen-type reduction

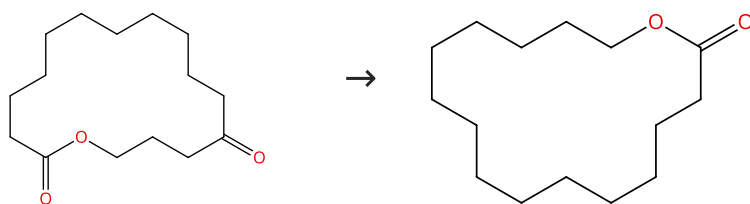
By: Salvador, J. A. R.; et al

Tetrahedron Letters (1993), 34(2), 361-2.

1.1 Reagents: Zinc
Solvents: Acetic acid

Scheme 19 (5 Reactions)

Steps: 1 Yield: 79-90%



Suppliers (5)

Suppliers (86)

31-538-CAS-1156102

Steps: 1 Yield: 90%

Simple method of reducing higher oxo lactones to lactones by a modified Clemmensen reaction with nonactivated zinc

By: Zakharkin, L. I.; et al

Zhurnal Organicheskoi Khimii (1987), 23(2), 296-8.

1.1 Reagents: Zinc, Hydrochloric acid
Solvents: Acetic anhydride

31-538-CAS-8151948

Steps: 1 Yield: 85%

Simple method of reducing higher oxo lactones to lactones by a modified Clemmensen reaction with nonactivated zinc

By: Zakharkin, L. I.; et al

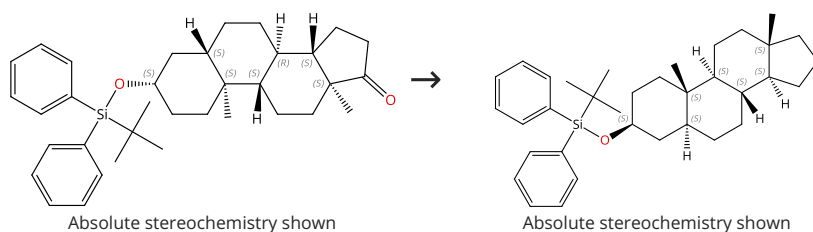
Zhurnal Organicheskoi Khimii (1987), 23(2), 296-8.

1.1 Reagents: Zinc, Hydrochloric acid
Solvents: Butyl ether

31-538-CAS-12406221 Steps: 1 Yield: 80% 1.1 Reagents: Zinc, Hydrochloric acid Solvents: Toluene	Simple method of reducing higher oxo lactones to lactones by a modified Clemmensen reaction with nonactivated zinc By: Zakharkin, L. I.; et al Zhurnal Organicheskoi Khimii (1987), 23(2), 296-8.
31-538-CAS-10278981 Steps: 1 Yield: 80% 1.1 Reagents: Zinc, Hydrochloric acid Solvents: Benzene	Simple method of reducing higher oxo lactones to lactones by a modified Clemmensen reaction with nonactivated zinc By: Zakharkin, L. I.; et al Zhurnal Organicheskoi Khimii (1987), 23(2), 296-8.
31-538-CAS-14535662 Steps: 1 Yield: 79% 1.1 Reagents: Zinc, Hydrochloric acid Solvents: Heptane	Simple method of reducing higher oxo lactones to lactones by a modified Clemmensen reaction with nonactivated zinc By: Zakharkin, L. I.; et al Zhurnal Organicheskoi Khimii (1987), 23(2), 296-8.

Scheme 20 (1 Reaction)

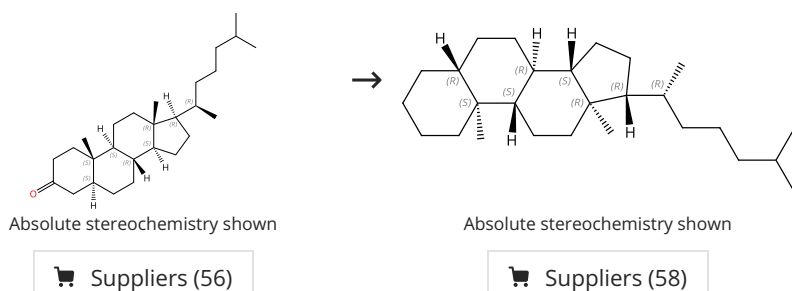
Steps: 1 Yield: 90%



31-538-CAS-15085888 Steps: 1 Yield: 90% 1.1 Reagents: Chlorotrimethylsilane, Zinc Solvents: Ethanol, Dichloromethane; 1 h, 0 °C 1.2 Reagents: Sodium bicarbonate; 5 min, 0 °C Experimental Protocols	One-pot reductive cleavage of exo-olefin to methylene with a mild ozonolysis-Clemmensen reduction sequence By: Xu, Shu; et al Tetrahedron Letters (2010), 51(34), 4534-4537.
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Scheme 21 (1 Reaction)

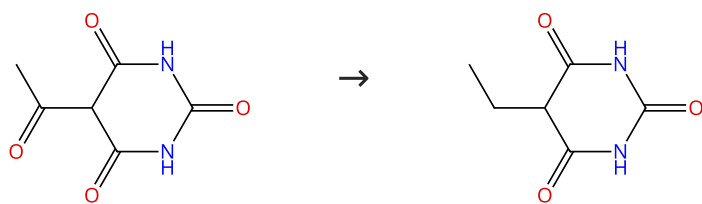
Steps: 1 Yield: 89%



31-538-CAS-15495064 Steps: 1 Yield: 89% 1.1 Reagents: Zinc Solvents: Acetic acid	Ultrasound assisted zinc reactions in synthesis. 2. A new Clemmensen-type reduction By: Salvador, J. A. R.; et al Tetrahedron Letters (1993), 34(2), 361-2.
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Scheme 22 (1 Reaction)

Steps: 1 Yield: 88%



Suppliers (26)

Suppliers (34)

31-538-CAS-15387094

Steps: 1 Yield: 88%

Mono C-alkylation and mono C-benylation of barbituric acids through zinc/acid reduction of acyl, benzylidene, and alkylidene barbiturate intermediates

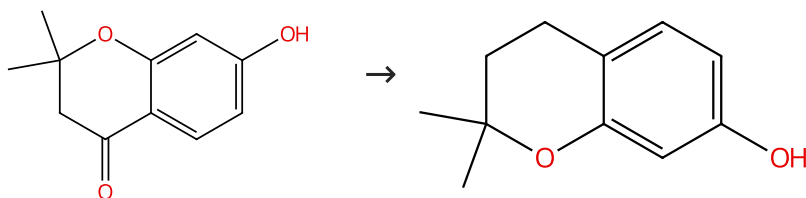
By: Jursic, Branko S.; et al

Tetrahedron Letters (2003), 44(10), 2203-2210.

1.1 Reagents: Zinc, Hydrochloric acid
Solvents: Water; 30 min, rt

Scheme 23 (1 Reaction)

Steps: 1 Yield: 88%



Suppliers (65)

Suppliers (32)

31-538-CAS-7324237

Steps: 1 Yield: 88%

Synthesis of 6-benzoyl-2,2-dimethyl-7-hydroxychroman

By: Fu, Juan; et al

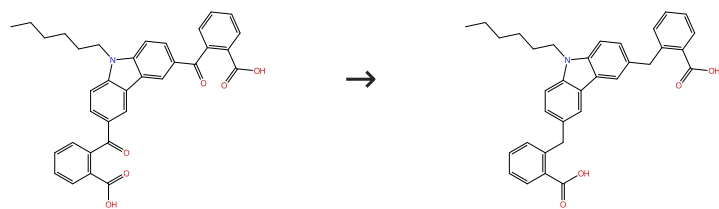
Jingxi Huagong Zhongjianti (2011), 41(4), 34-36.

1.1 Reagents: Mercury alloy, nonbase, Hg, Zn
Solvents: Ethanol; rt → reflux

1.2 Reagents: Hydrochloric acid
Solvents: Water; reflux

Scheme 24 (1 Reaction)

Steps: 1 Yield: 88%



31-538-CAS-3914303

Steps: 1 Yield: 88%

Heterocyclic heptacene analogs - 8H-16,17-epoxydinaphtho[2,3-c:2',3'-g]carbazoles as charge transport materials

By: Rimkus, Renaldas; et al

Dyes and Pigments (2016), 124, 133-144.

1.1 Reagents: Zinc, Mercury dibromide
Solvents: Water

1.2 Solvents: 1,4-Dioxane; 4 - 24 h, 50 °C

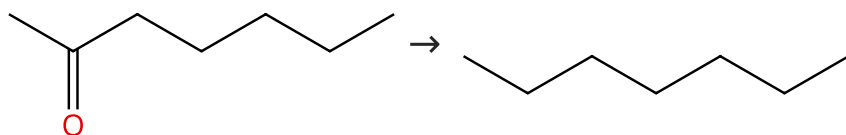
1.3 Reagents: Hydrochloric acid
Solvents: Water; 4 h

1.4 Reagents: Water; 12 h, heated

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 88%



Suppliers (68)

Suppliers (245)

31-538-CAS-13472441

Steps: 1 Yield: 88%

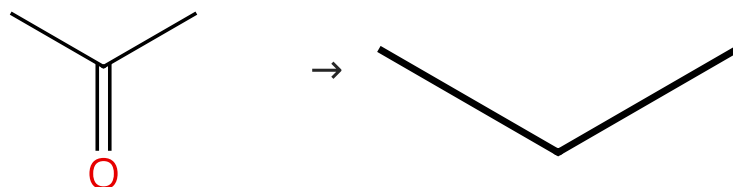
Clemmensen reductions using ultrasonic irradiation

1.1 **Reagents:** Hydrogen iodide, Mercury alloy, nonbase, Hg, Zn
Solvents: Methanol, Water

By: Reeves, W. Preston; et al
 Synthetic Communications (1988), 18(16-17), 1961-6.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 88%



Suppliers (296)

Suppliers (13)

31-538-CAS-3931640

Steps: 1 Yield: 88%

Clemmensen reduction of difunctional ketones

1.1 **Reagents:** Hydrazine

By: Buchanan, J. G. St. C.; et al
 Quarterly Reviews, Chemical Society (1969), 23(4), 522-36.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 87%



31-538-CAS-9449963

Steps: 1 Yield: 87%

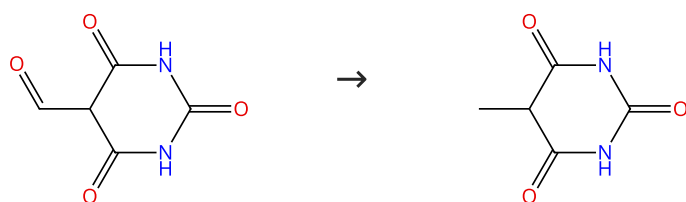
Synthesis and characterization of some 6,6'-substituted derivatives of 2,2-bis(ferrocenyl)propane

1.1 **Reagents:** Zinc, Mercuric chloride
Solvents: Methanol, Benzene; reflux
 1.2 **Reagents:** Hydrochloric acid
Solvents: Methanol; reflux; 5 h, reflux

By: Wang, Xiao-li; et al
 Huaxue Shiji (2009), 31(11), 924-926.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 87%



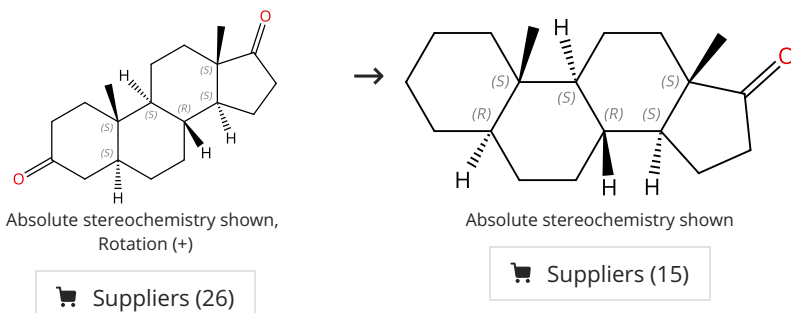
Suppliers (14)

Suppliers (36)

31-538-CAS-6862038 Steps: 1 Yield: 87% 1.1 Reagents: Zinc, Hydrochloric acid Solvents: Water; 30 min, rt	Mono C-alkylation and mono C-benylation of barbituric acids through zinc/acid reduction of acyl, benzylidene, and alkylidene barbiturate intermediates By: Jursic, Branko S.; et al Tetrahedron Letters (2003), 44(10), 2203-2210.
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Scheme 29 (1 Reaction)

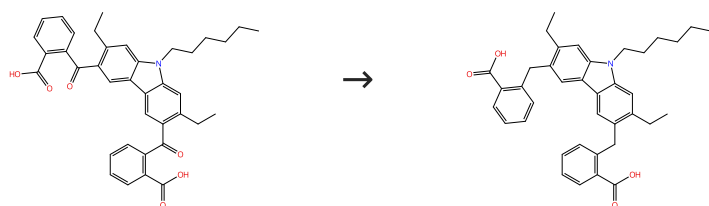
Steps: 1 Yield: 87%



31-538-CAS-4229133 Steps: 1 Yield: 87% 1.1 Reagents: Zinc Solvents: Acetic acid	Ultrasound assisted zinc reactions in synthesis. 2. A new Clemmensen-type reduction By: Salvador, J. A. R.; et al Tetrahedron Letters (1993), 34(2), 361-2.
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Scheme 30 (1 Reaction)

Steps: 1 Yield: 87%



31-538-CAS-12151446 Steps: 1 Yield: 87% 1.1 Reagents: Zinc, Mercury dibromide Solvents: Water 1.2 Solvents: 1,4-Dioxane; 4 - 24 h, 50 °C 1.3 Reagents: Hydrochloric acid Solvents: Water; 4 h 1.4 Reagents: Water; 12 h, heated Experimental Protocols	Heterocyclic heptacene analogs - 8H-16,17-epoxydinaphto[2,3-c:2',3'-g]carbazoles as charge transport materials By: Rimkus, Renaldas; et al Dyes and Pigments (2016), 124, 133-144.
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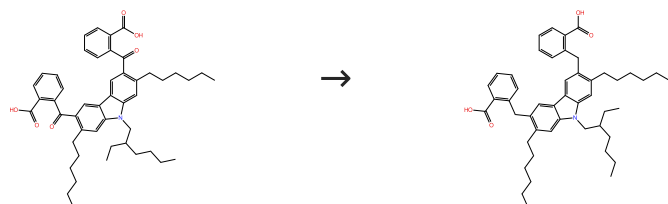
Scheme 31 (1 Reaction)

Steps: 1 Yield: 86%



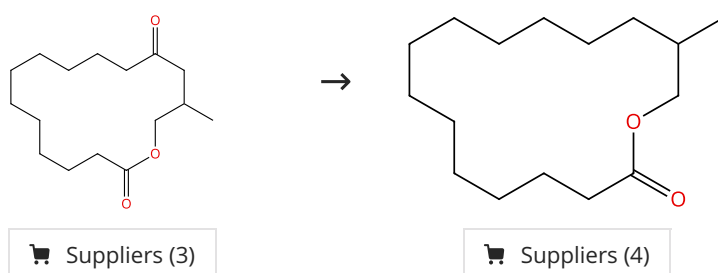
31-538-CAS-10440694 Steps: 1 Yield: 86% 1.1 Reagents: Zinc, Mercuric chloride Solvents: Methanol, Benzene; rt 1.2 Reagents: Hydrochloric acid Solvents: Methanol; rt; 5 h, rt	Synthesis and characterization of 6-propionyl-6'-butyryl-(2,2-bisferrocenylpropane) and its reduction products By: Wang, Xiao-li; et al Neimenggu Shifan Daxue Xuebao, Ziran Kexue Hanwenban (2009), 38(3), 296-298, 303.
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Scheme 32 (1 Reaction)

Steps: **1** Yield: **86%**

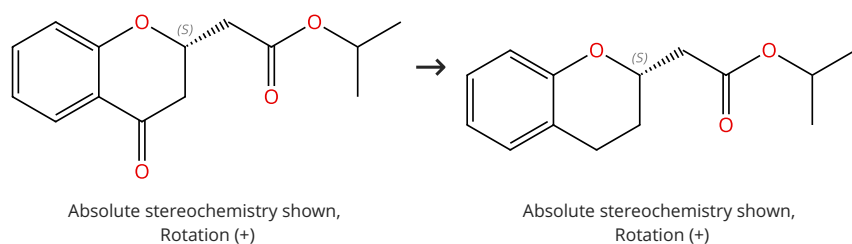
31-538-CAS-10240189 Steps: 1 Yield: 86% 1.1 Reagents: Zinc, Mercury dibromide Solvents: Water 1.2 Solvents: 1,4-Dioxane; 4 - 24 h, 50 °C 1.3 Reagents: Hydrochloric acid Solvents: Water; 4 h 1.4 Reagents: Water; 12 h, heated Experimental Protocols	Heterocyclic heptacene analogs - 8H-16,17-epoxydinaphtho[2,3-c:2',3'-g]carbazoles as charge transport materials By: Rimkus, Renaldas; et al Dyes and Pigments (2016), 124, 133-144.
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Scheme 33 (1 Reaction)

Steps: **1** Yield: **85%**

31-538-CAS-5393477 Steps: 1 Yield: 85% 1.1 Reagents: Zinc, Hydrochloric acid Solvents: Acetic anhydride	Simple method of reducing higher oxo lactones to lactones by a modified Clemmensen reaction with nonactivated zinc By: Zakharkin, L. I.; et al Zhurnal Organicheskoi Khimii (1987), 23(2), 296-8.
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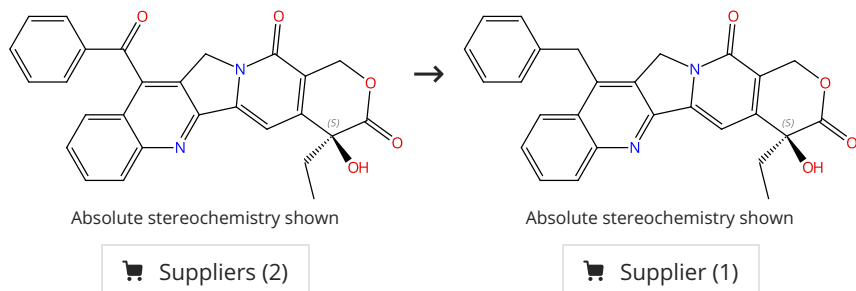
Scheme 34 (1 Reaction)

Steps: **1** Yield: **85%**

31-538-CAS-19750588 Steps: 1 Yield: 85%	Helical Multi-Coordination Anion-Binding Catalysts for the Highly Enantioselective Dearomatization of Perylene Derivatives
1.1 Reagents: Acetic acid, Zinc; rt; 18 h, 100 °C Experimental Protocols	By: Fischer, Theresa; et al Angewandte Chemie, International Edition (2019), 58(10), 3217-3221.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 85%



31-538-CAS-5739180 Steps: 1 Yield: 85%	Study on synthesis of 7-benzyl-camptothecin
1.1 Reagents: Hydrochloric acid, Ferrous sulfate Solvents: Dichloromethane, Water; 1.5 h, reflux	By: Guo, Wei; et al Huaxue Shiji (2008), 30(2), 141-142, 144.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 84%



31-538-CAS-3846543 Steps: 1 Yield: 84%	Heterocyclic heptacene analogs - 8H-16,17-epoxydinaphtho[2,3-c:2',3'-g]carbazoles as charge transport materials
1.1 Reagents: Zinc, Mercury dibromide Solvents: Water 1.2 Solvents: 1,4-Dioxane; 4 - 24 h, 50 °C 1.3 Reagents: Hydrochloric acid Solvents: Water; 4 h 1.4 Reagents: Water; 12 h, heated Experimental Protocols	By: Rimkus, Renaldas; et al Dyes and Pigments (2016), 124, 133-144.

Scheme 37 (1 Reaction)

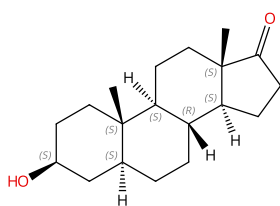
Steps: 1 Yield: 83%



31-538-CAS-7304852 Steps: 1 Yield: 83%	Synthesis and characterization of some 6,6'-substituted derivatives of 2,2-bis(ferrocenyl)propane
1.1 Reagents: Zinc, Mercuric chloride Solvents: Methanol, Benzene; reflux 1.2 Reagents: Hydrochloric acid Solvents: Methanol; reflux; 5 h, reflux	By: Wang, Xiao-li; et al Huaxue Shiji (2009), 31(11), 924-926.

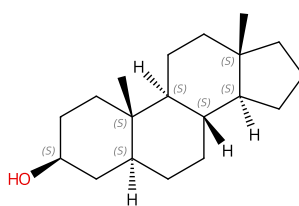
Scheme 38 (1 Reaction)

Steps: 1 Yield: 82%



Absolute stereochemistry shown

Suppliers (37)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (24)

31-538-CAS-8704038

Steps: 1 Yield: 82%

One-pot reductive cleavage of exo-olefin to methylene with a mild ozonolysis-Clemmensen reduction sequence

- 1.1 Reagents: Chlorotrimethylsilane, Zinc
Solvents: Methanol, Dichloromethane; 1 h, 0 °C
- 1.2 Reagents: Sodium bicarbonate; 5 min, 0 °C

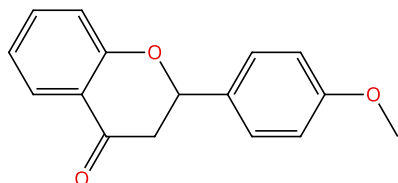
By: Xu, Shu; et al

Tetrahedron Letters (2010), 51(34), 4534-4537.

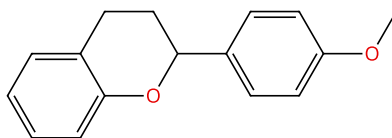
Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 82%



Suppliers (40)



Suppliers (4)

31-538-CAS-12993293

Steps: 1 Yield: 82%

Reduction of some flavanones and E-3-benzylideneflavanones under modified Clemmensen reduction condition

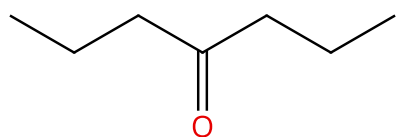
- 1.1 Reagents: Acetic acid, Mercury alloy, nonbase, Hg, Zn
Solvents: Ethanol, Water; rt; 1 - 1.5 h, reflux

By: Dey, Sankar P.; et al

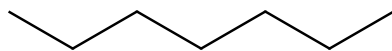
Journal of the Indian Chemical Society (2008), 85(7), 717-720.

Scheme 40 (1 Reaction)

Steps: 1 Yield: 82%



Suppliers (62)



Suppliers (245)

31-538-CAS-7349938

Steps: 1 Yield: 82%

Clemmensen reductions using ultrasonic irradiation

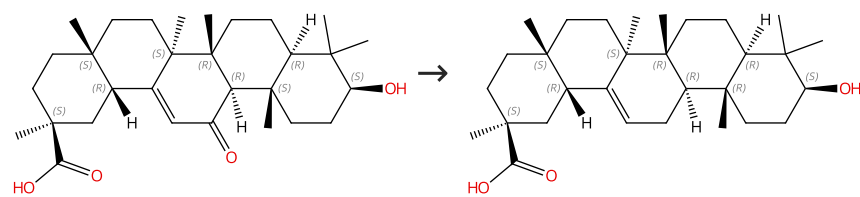
- 1.1 Reagents: Hydrogen iodide, Mercury alloy, nonbase, Hg, Zn
Solvents: Methanol, Water

By: Reeves, W. Preston; et al

Synthetic Communications (1988), 18(16-17), 1961-6.

Scheme 41 (2 Reactions)

Steps: 1 Yield: 77-81%



Absolute stereochemistry shown

Suppliers (103)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (14)

31-538-CAS-4491581

Steps: 1 Yield: 81%

**Synthesis and theoretical study on the structure of 11-deoxy
Iglycyrhretinic acid**

By: Yong, Jianping

Jisuanji Yu Yingyong Huaxue (2011), 28(10), 1325-1329.

1.1 Reagents: Zinc, Mercuric chloride, Hydrochloric acid

Solvents: Water; 30 min, rt

1.2 Reagents: Hydrochloric acid

Solvents: 1,4-Dioxane, Water; cooled; 30 min, rt

31-538-CAS-1248411

Steps: 1 Yield: 77%

Synthesis of glycyrrhetic acid derivatives

By: Yong, J.-P.; et al

Chemistry of Natural Compounds (2008), 44(2), 194-196.

1.1 Reagents: Zinc, Mercuric chloride, Hydrochloric acid

Solvents: Water; rt; 30 min, rt

1.2 Reagents: Hydrochloric acid

Solvents: 1,4-Dioxane, Water; 30 min, cooled; 30 min, cooled

Experimental Protocols

Scheme 42 (1 Reaction)

Steps: 1 Yield: 80%



31-538-CAS-15105242

Steps: 1 Yield: 80%

**Heterocyclic heptacene analogs - 8H-16,17-epoxydinaphtho[2,
3-c:2',3'-g]carbazoles as charge transport materials**

By: Rimkus, Renaldas; et al

Dyes and Pigments (2016), 124, 133-144.

1.1 Reagents: Zinc, Mercury dibromide

Solvents: Water

1.2 Solvents: 1,4-Dioxane; 4 - 24 h, 50 °C

1.3 Reagents: Hydrochloric acid

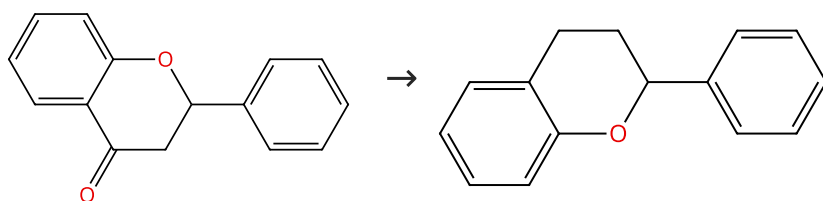
Solvents: Water; 4 h

1.4 Reagents: Water; 12 h, heated

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 79%

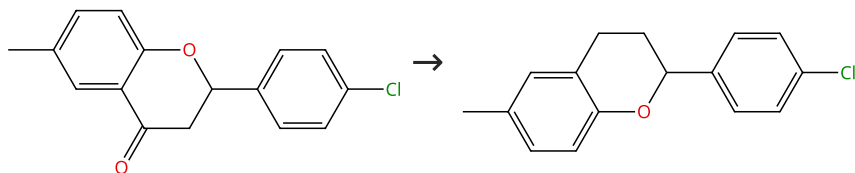


Suppliers (85)

Suppliers (21)

31-538-CAS-8325684 Steps: 1 Yield: 79%	Reduction of some flavanones and E-3-benzylidene flavanones under modified Clemmensen reduction condition By: Dey, Sankar P.; et al Journal of the Indian Chemical Society (2008), 85(7), 717-720.
1.1 Reagents: Acetic acid, Mercury alloy, nonbase, Hg,Zn Solvents: Ethanol, Water; rt; 1 - 1.5 h, reflux	

Scheme 44 (1 Reaction)

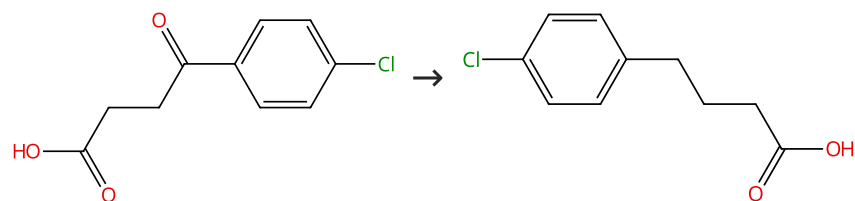
Steps: **1** Yield: **78%**

Suppliers (6)

Suppliers (2)

31-538-CAS-6873059 Steps: 1 Yield: 78%	Reduction of some flavanones and E-3-benzylidene flavanones under modified Clemmensen reduction condition By: Dey, Sankar P.; et al Journal of the Indian Chemical Society (2008), 85(7), 717-720.
1.1 Reagents: Acetic acid, Mercury alloy, nonbase, Hg,Zn Solvents: Ethanol, Water; rt; 1 - 1.5 h, reflux	

Scheme 45 (1 Reaction)

Steps: **1** Yield: **78%**

Suppliers (91)

Suppliers (70)

31-538-CAS-17800646 Steps: 1 Yield: 78%	Synthesis of tetralone compounds By: Jin, Hao; et al Lanzhou Ligong Daxue Xuebao (2013), 39(4), 70-73.
1.1 Reagents: Sodium hydroxide, Hydrazine hydrate (1:1) Solvents: Triethylene glycol; 3 h, reflux; 1 h, 210 °C; 210 °C → 190 °C; 4 h, 190 °C; 190 °C → rt	
1.2 Reagents: Hydrochloric acid Solvents: Water; pH 1	